

Indian Institute of Technology
Department of Management Studies

MS6015: Multivariate Statistical Methods for Business
Sample PCA, Factor and Cluster

April 30, 2026

PCA and Factor analysis

1. A researcher obtained seven physical characteristics (denoted by $X_1, \dots, X_7$) for each of 3000 students. The sample correlation matrix is given below.

	X_1	X_2	X_3	X_4	X_5	X_6	X_7
X_1	1	0.402	0.395	0.301	0.305	0.399	0.340
X_2	0.402	1	0.618	0.150	0.135	0.206	0.183
X_3	0.395	0.618	1	0.321	0.289	0.363	0.345
X_4	0.301	0.150	0.321	1	0.846	0.759	0.661
X_5	0.305	0.135	0.289	0.846	1	0.797	0.800
X_6	0.399	0.206	0.363	0.759	0.797	1	0.736
X_7	0.340	0.183	0.345	0.661	0.800	0.736	1

The eigen vectors and eigen values of the correlation matrix are

	1	2	3	4	5	6	7
Eigenvectors	.285	-.351	.877	-.088	-.076	.112	-.023
	.211	-.643	-.246	.686	-.098	-.010	.020
	.294	-.515	-.387	-.693	-.112	.029	-.074
	.435	.240	-.113	.126	-.604	.330	.500
	.453	.282	-.079	.127	-.024	.270	-.787
	.453	.167	.028	.023	-.065	-.873	.024
	.434	.182	-.027	-.090	.776	.208	.352
Eigenvalues	3.82	1.49	0.65	0.36	0.34	0.23	0.11

Answer the following questions:

[5 × 2 = 10 marks]

- (a) What is the percentage of the total standardized variation attributed to the second principal component?

(1.49/7) × 100% = 21.3%

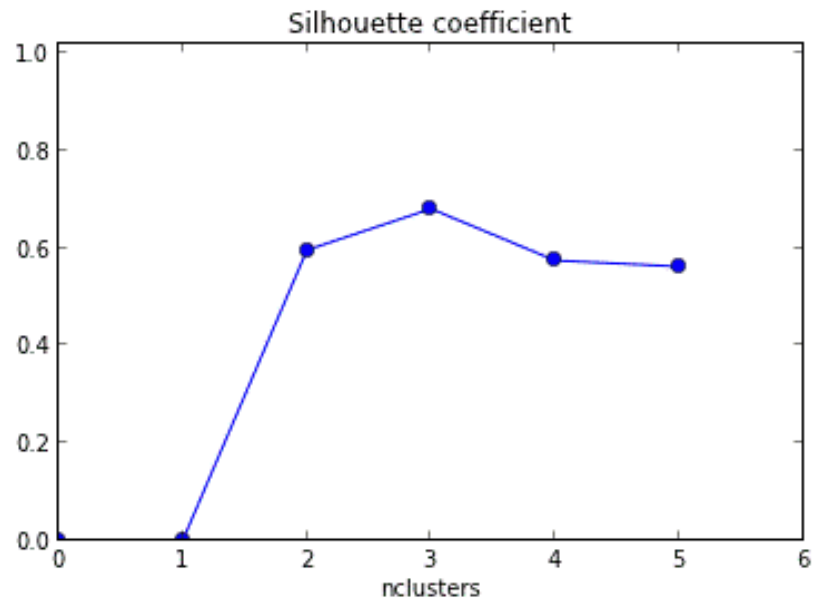
- (b) Estimate the correlation between the scores for the first and second principal components.
0
- (c) Estimate the variance of the scores for the first principal component.
3.82
- (d) Give a brief interpretation of the first principal component.
This is an overall size component with more emphasis on body length and height measurements than on head size measurements. This component accounts for about 54.6% of the total variance of the standardized measurements.
- (e) Describe how the first three principal components are related to patterns in the correlation matrix.
The first principal component corresponds to the positive correlations between all of the measurements. The higher loadings of the body length measurements on the first component reflects that correlations among the body length measurements are stronger than 1 correlations among the head size measurements. The second principal component corresponds to the tendency for correlations between head size and body length measurements to be weaker than either the correlations among the head size measurements or the correlations among the body length measurements. The third principal component reflects weaker correlations between head length and head width measurements than the correlation between the head width measurements.

Clustering

2. Answer the following two questions using the following information: Assume, you want to cluster 7 observations into 3 clusters using K-Means clustering algorithm. After first iteration clusters, C1, C2, C3 has following observations:
C1: {(2,2), (4,4), (6,6)}
C2: {(0,4), (4,0)}
C3: {(5,5), (9,9)}

What will be the cluster centroids if you want to proceed for second iteration?

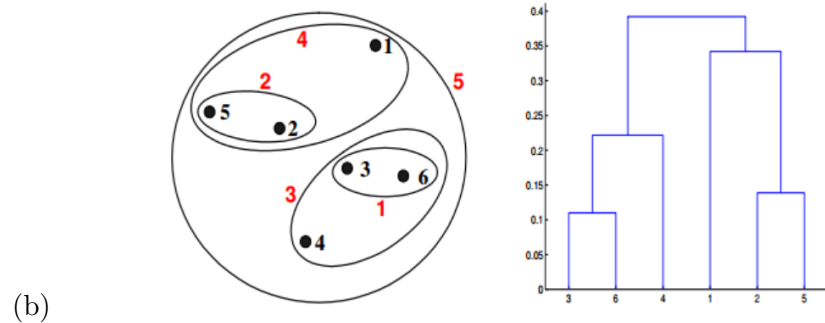
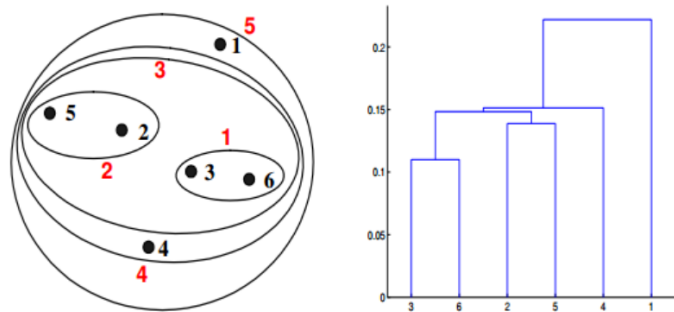
- (a) C1: (4,4), C2: (2,2), C3: (7,7)
(b) C1: (6,6), C2: (4,4), C3: (9,9)
(c) C1: (2,2), C2: (0,0), C3: (5,5)
(d) None of the choices
3. What will be the Manhattan distance for observation (9, 9) from cluster centroid C1 in second iteration.
(a) 10
(b) $5 * \sqrt{2}$
(c) $13 * \sqrt{2}$
(d) None of the choices
4. What should be the best choice of no. of clusters based on the following results:



- (a) 1
- (b) 2
- (c) 3
- (d) 4

5. Which of the following clustering representations and dendrogram depicts the use of Single linkage proximity function in hierarchical clustering:

- (a) [Right answer](#)



(b)

